

Chapter 39 — Principles of Laparoscopic and Robotic Surgery in Children

Campbell-Walsh-Wein Urology · Part V Paediatric Urology · Chapter Notes

HISTORY OF MINIMALLY INVASIVE SURGERY IN CHILDREN

- First paediatric urological laparoscopy was **diagnostic, for the non-palpable testis** (1976): it confirmed presence or absence of the testis, avoided open exploration and changed the management of cryptorchidism. Initially resisted, it **irreversibly changed** paediatric urology.
- 1990s, **extirpative surgery**: laparoscopic nephrectomy was established in adults, then reported in children (1994). Paediatric uptake was slower because of port size and ergonomics.
- **Laparoscopic pyeloplasty was the first reconstructive minimally invasive surgery (MIS)** in children (1995). Early cases were long because laparoscopic suturing is difficult; as exposure grew, adoption widened and indications expanded.
- The **da Vinci platform was approved for adult urology in 2000; the first paediatric robotic procedure followed in 2002**. It corrected the ergonomic and technical difficulties of pure laparoscopy, and robotic **pyeloplasty and ureteric reimplantation** are now routine, with increasingly complex reconstruction.
- **Sequence**: diagnostic (1976) → extirpative (1994) → reconstructive (1995) → robotic (2002).

Concept — Each stage was gated by the same bottleneck, intracorporeal suturing and ergonomics; the robot removed that bottleneck, which is why reconstruction, not extirpation, is where paediatric robotics has grown.

ACCESS

- Insufflation needs **initial peritoneal entry, by two classic approaches**.

OPEN VERSUS CLOSED ACCESS

	Open access (Hasson)	Closed access (Veress needle)
Principle	Entry under direct vision ; port placed before insufflation	Blind entry with a spring-loaded blunt-tipped needle; insufflate before port placement
Steps	Skin incision (about port size) to fascia → fascia and peritoneum opened under vision → port inserted → insufflation	Needle passed, clicking through each layer → aspirate → saline test → insufflate → check opening pressure
Rationale	Ensures intraperitoneal port position; avoids intra-abdominal injury	Position confirmed by tests before gas is given
Origin	—	Designed by János Veres (1938) for therapeutic pneumothorax in tuberculosis ; first used for insufflation by Raoul Palmer, gynaecologist (1947)
Evidence	Cochrane review: no difference in successful entry, mortality, vascular or visceral injury. Children: no robust comparisons; both safe and reproducible	

Veress technique

- **Site**: the **umbilicus** in children. If entry fails or prior scarring raises concern, use **Palmer's point**: left upper quadrant, midclavicular line, about **3 fingerbreadths below the costal margin**.
- **Confirmation: aspirate first**, to exclude a vessel or hollow viscus → inject saline, which should **drop immediately**; if it does not, the needle is malpositioned or the patient insufficiently relaxed → connect insufflation and **monitor opening pressure**.

- **Right common iliac vein** injury is reported with Veress entry because it lies near the umbilicus.

KEY POINTS: ACCESS

- **Open (Hasson)** access gives direct visualisation and port placement into the peritoneum before insufflation.
- **Closed (Veress)** access uses a spring-loaded needle to enter and insufflate before port placement.
- The **right common iliac vein** is the vessel most commonly injured during Veress placement.
- In children there are **no data showing superiority** or lower risk of either modality.

Concept — The two techniques trade direct vision against speed, yet outcomes do not differ; what protects the patient is **confirmation of intraperitoneal position** (aspiration, saline drop, low opening pressure), not the choice of method.

PATIENT POSITIONING

- Positioning determines **patient safety, robotic system optimisation and surgeon ergonomics**.

RENAL VERSUS PELVIC POSITIONING

	Transperitoneal renal	Pelvic
Position	Modified flank , at the bed edge (full instrument range, no clash with bed)	Supine ; frog-leg if urethral or genital access is needed
Exposure	Flank elevated 30–45° on a jelly roll so bowel falls away from the retroperitoneum	Steep Trendelenburg
Limbs	Lower leg bent, upper leg straight , pillows or foam between; shoulder padded; lower arm on armboard out of robotic arm range	Pressure points padded
Security	Belts and tape, no tape over joints or directly on skin; test table rotation before preparation	Strapped securely for steep tilt; a metal shelf ("head butler") protects the head from instruments

- **Transmesenteric renal cases** need less flank elevation and rotation because the **colon is not reflected**.
- **Musculoskeletal anomalies** — contracture, limited range of motion, scoliosis, common in spina bifida — need thoughtful positioning and generous padding.

Concept — Positioning uses **gravity to create exposure and fixation to survive table tilt**; everything else is protection of pressure points and skin.

PHYSIOLOGY OF PNEUMOPERITONEUM

- Working space is made by filling potential space with **inert gas**; raised **intra-abdominal pressure (IAP)** causes changes absent in open surgery.
- Routine MIS raises IAP to **12–15 mm Hg**, with haemodynamic, intracranial, pulmonary and renal effects. In children these are minimised by keeping **IAP below 12 mm Hg**, and **lower pressures (about 7 mm Hg)** are **feasible and safe**.

Cardiovascular Changes

- **Insufflation above 10 mm Hg decreases venous return** and so cardiac index; **lower pressures increase venous return** and cardiac index.
- Catecholamine release with **renin-angiotensin-aldosterone system (RAAS)** activation causes vasoconstriction, raising **mean arterial pressure** and **systemic vascular resistance**.

- **Carbon dioxide (CO₂) absorption** causes hypercarbia and acidosis, which **vasodilate and depress contractility** — opposing the RAAS effect.
- Initial peritoneal distension can provoke either reflex:

VAGAL VERSUS SYMPATHETIC RESPONSE TO INSUFFLATION

Vagal	Sympathetic
Cardiovascular instability and bradycardia	Tachycardia

- **Management:** tell the anaesthetic team when insufflation starts → if unstable, **desufflate, then reinsufflate slowly** → most procedures can then continue.

Pulmonary Changes

- Pressure on the diaphragm **reduces pulmonary excursion**; cephalad lung displacement **raises peak airway pressure**.
- **CO₂ is the gas of choice** — non-combustible, soluble and excreted by respiration. Greater absorption needs **higher minute ventilation**; with lung disease it can cause **hypercarbia, arrhythmia and acidosis**.
- **Infants and small children** are affected most: IAP up to 15 mm Hg **reduces tidal volume and compliance**. **Trendelenburg** for pelvic surgery worsens this, especially with cardiopulmonary comorbidity.

Intracranial Pressure

- **Hypercarbia raises cerebral blood flow**, and intracranial blood flow can rise **even when CO₂ is controlled** (mechanism unclear) — relevant to **prolonged Trendelenburg** and **ventriculoperitoneal (VP) shunts**.

VENTRICULOPERITONEAL SHUNTS: HISTORICAL CONCERN VERSUS CURRENT EVIDENCE

Historical concern	Current evidence
Raised intracranial pressure (ICP) : shunt obstruction by the pneumoperitoneum plus venous outflow obstruction from raised intrathoracic and intra-abdominal pressure	MIS with VP shunts is generally safe, without significant ICP rise , even at elevated pressures

Renal Effects

- Raised IAP reduces **renal perfusion and oxygenation**. **Rat**: IAP of 15 mm Hg or more significantly reduces renal perfusion, with an **inverse IAP–oxygenation relationship**. **Swine renal-insufficiency model** (20 mm Hg for 6 hours): reduced **glomerular filtration rate (GFR)**, urine output and renal blood flow immediately, but GFR at baseline by 1 week — the effect is **temporary**.
- **Children, flank near-infrared spectroscopy**: age-appropriate pressures **do not significantly reduce** renal perfusion or urine output.

AGE-APPROPRIATE MAXIMUM INSUFFLATION PRESSURES

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	≤ 6 mm Hg	≤ 8 mm Hg	≤ 10 mm Hg	≤ 12 mm Hg

- Pattern: **2 mm Hg per age step, from 6 to 12**; the oldest band equals the general paediatric ceiling.
- **Intraoperative and early postoperative oliguria** is most pronounced **under 1 year**, from reduced renal perfusion. It is **reversible**, and urine output during pneumoperitoneum is **not a reliable guide to fluid resuscitation**.

KEY POINTS: PHYSIOLOGY OF THE PNEUMOPERITONEUM

- Insufflation pressures **even below 10 mm Hg** may affect cardiac preload and afterload, with implications in **complex cardiac anatomy**.

- **Intracranial pressure may increase** with pneumoperitoneum.
- Oliguria from pneumoperitoneum should **not be treated initially with intravenous fluid**.

Concept — Almost every effect is **pressure-dependent and transient**: dosed by IAP and reversed by desufflation, so the paediatric strategy is the lowest age-appropriate pressure rather than treating consequences — hence no fluid bolus for oliguria.

EQUIPMENT

- Ports and instruments have evolved under **constant demand for smaller technology** for children.

Ports

REUSABLE VERSUS DISPOSABLE PORTS

Reusable	Disposable
Metal; perforator tip blunt or sharp	Plastic; perforator tip bladed or blunt

- Diameters **3, 5, 8, 10, 12 and 15 mm**. 3- and 5-mm laparoscopic ports suit very small children; most robotic platforms use **8 mm**.
- **Radially expanding access (REA) cannulas** dilate from **Veress diameter up to 12 mm**, designed to reduce port-site pain and potentially hernia and bleeding. They show **less port-site bleeding**; evidence on other complications is lacking.

VALVED VERSUS VALVELESS TROCARS

Valved (traditional)	Valveless
One-way valve; pressure drops when instruments pass or suction runs	Recirculates gas : constant flow and pressure; safe, less CO₂ ; subcutaneous emphysema at the trocar reported

Port Placement

- Placement follows the **number of instruments** and **access to the target**; there are **no formal rules** — preference and habitus decide.

TRIANGULATION VERSUS LINEAR PORT PLACEMENT

Triangulation	Linear
Traditional; essential in pure laparoscopy; optimal ergonomics, less clashing	Preferred by many in robotic paediatric cases; more working space, fewer collisions ; feasible on the da Vinci Xi

- **"Burping" the ports**: clutch the robotic port and pull upwards on the skin to enlarge the working space.
- **Hidden incision endoscopic surgery (HIdES)** (described 2011) **moves the camera port** so that ports lie **below the Pfannenstiel line**. Used for renal surgery, ureteric reimplantation, urachal remnant excision and **single-port surgery**; **equivalent outcomes** for robotic pyeloplasty in small series. Advantage: **no visible scar above the clothing line**; yet **only a minority** of surveyed paediatric urologists use it.

KEY POINTS: PORT PLACEMENT

- Pure laparoscopy relies on **triangulation** for optimal ergonomics and instrument efficiency.
- Modern robotic platforms allow **more freedom in port position**.
- Placement variations allow ports in **less visible areas**.

Instruments

- Pure laparoscopic instruments mimic open ones but offer only **4 degrees of freedom (DOF)**: in/out, up/down, left/right and axial rotation. They are sized to ports; **5- and 3-mm instruments** suit small children.

ATRAUMATIC VERSUS TRAUMATIC GRASPERS

Atraumatic	Traumatic
Fine serrations ; bowel, delicate tissue	Deep serrations or teeth ; secure grip

- Cutting instruments mirror open ones; many connect to cautery with an **insulated sheath** against thermal injury.

LAPAROSCOPIC VERSUS ROBOTIC NEEDLE DRIVERS

Laparoscopic	Robotic
Locking ; straight or curved (right and left versions); not wristed — complex suturing hard	Various sizes; straight tip only ; wristed — suturing easier

Haemostatic energy devices

ENERGY MODALITIES COMPARED

	Monopolar	Bipolar (including LigaSure)	Harmonic
Energy	Electrical	Electrical	Ultrasonic
Circuit	Patient grounded ; current passes instrument → tissue	No grounding ; current passes through tissue between the jaws	No electrical circuit
Sealing	—	LigaSure seals vessels up to 7 mm , divides tissue; three energy settings by desiccation level	Seals up to 7 mm
Risk	Any exposed metal conducts → thermal injury; insulation added	Less collateral damage than monopolar	Equivalent to bipolar
Fits	Most laparoscopic and robotic instruments, including scissors and graspers	Dedicated devices	Dedicated devices

- **Harmonic versus bipolar sealers** (non-urological studies): no significant difference in blood loss, adjacent tissue damage or postoperative bleeding — **vendor or surgeon choice**.

Robotic Technology

- First human robotic trial: **1997, "Mona"** (after the Mona Lisa; Intuitive Surgical). The bedside assistant held the camera, but **wristed, exchangeable instruments preserved the sterile field** — the first generation.
- **da Vinci is the most widely adopted**, with up to **four independent arms**.

DA VINCI XI VERSUS DA VINCI SP

Xi (multiport)	SP (single port)
Ports as small as 8 mm ; no 5-mm ports	One 25-mm port : camera + three double-jointed wristed instruments

Concept — Equipment evolution is driven by **miniaturisation for small bodies and restoration of the wrist lost in laparoscopy**; the robot restores the wrist at a larger minimum size (8-mm ports, 25-mm single port), which is the central paediatric tension.

PATIENT SELECTION

- Individual **risk–benefit** assessment, including the **surgeon's ability and experience**; **anaesthetic risk and overall health** are usually the primary drivers.
- Significant **cardiac or pulmonary pathology**: evaluation by the primary physician and discussion with the anaesthetist about tolerance of pneumoperitoneum.
- **Learning-curve progression**: healthy children → unoperated abdomens → larger children → only then complex procedures in medically complex patients.

KEY POINTS: PATIENT SELECTION

- The **only strict contraindication** to MIS is inability to tolerate pneumoperitoneum.
- Significant **cardiopulmonary comorbidity** requires preoperative evaluation and optimisation with other specialists and anaesthetists.

Concept — Selection is **physiological, not anatomical**: tolerance of pneumoperitoneum decides candidacy, not age, size or diagnosis.

INFANT ROBOTIC SURGERY

- **Age and size are not contraindications**; the infant problem is **working space** — about **1 L at 1 year versus 6 L in an adult**.
- Robotic adoption is greatest in **adolescents**; **most infant pyeloplasties remain open**, though robotic use is rising.
- **Visualisation aids**: limit bowel distension by **minimising inhalational anaesthetic**; **preoperative enema**; **intraoperative rectal tube** to decompress flatus.
- Infant robotic pyeloplasty is **safe, feasible and equivalent**; against matched open cohorts it gives **shorter operative time and stay and less narcotic**. Deterrents: small workspace, **arm collisions**, reduced range of motion, learning curve.

KEY POINTS: INFANT ROBOTIC SURGERY

- **Minimising inhalational anaesthetic** decreases bowel distension and aids visualisation in the small abdomen.

Concept — Every infant difficulty follows from one number — **a working volume about a sixth of the adult's** — so every adjunct is an attempt to recover space.

SURGICAL TECHNIQUE

TRANSPERITONEAL, RETROPERITONEAL AND VESICOSCOPIC APPROACHES

	Transperitoneal	Retroperitoneal	Vesicoscopic
Route	Peritoneal cavity → retroperitoneum, incising one peritoneal layer	Directly into the retroperitoneum	CO ₂ -insufflated bladder
Space	Largest	Limited	Limited; smaller bladder predicts complications

	Transperitoneal	Retroperitoneal	Vesicoscopic
Strengths	Commonest ; familiar (shared with general surgery); shows adjacent organs and bowel for complex reconstruction	Avoids the peritoneum , lower morbidity	Avoids the peritoneum; intravesical access
Limitations	Colon must be mobilised or a window used	Unfamiliar view; peritoneal tear loses the pneumoretroperitoneum	Significant learning curve ; difficult access view; peritoneal violation at port placement; complex port-site closure
Uses	Intra-abdominal testis, bladder, lower ureter, kidney	Pyeloplasty, nephrectomy, heminephrectomy; lymph node dissection	Cross-trigonal ureteric reimplantation

Transperitoneal Approach

TRANSPERITONEAL RENAL ACCESS: RIGHT VERSUS LEFT

Right	Left
Right colon mobilised ; retrocolic approach	Ureter and pelvis sometimes via a mesenteric window ; minimal colonic handling

- In adults, **transmesenteric pyeloplasty shortened operative time and stay with equivalent function**; it is considered **advantageous in children when feasible**.

Retroperitoneal Approach

- Traditional for open urology. Versus open renal surgery: **shorter stay, equivalent outcomes** — useful in **peritoneal dialysis**, avoiding dialysis delay from peritoneal violation.
- Systematic review of transperitoneal versus retroperitoneal laparoscopic nephrectomy: **no significant advantage to either**; major complications rare. Used selectively; **specific mentorship** advised.
- Retroperitoneal **lymph node dissection** is reported for **paratesticular rhabdomyosarcoma** and **non-seminomatous germ cell tumours**.

Vesicoscopic Approach

- **First described 2005** for cross-trigonal reimplantation: **5-mm camera port through the bladder dome under cystoscopic vision** → two small working ports through the bladder wall → urethra occluded → bladder insufflated. Outcomes **equivalent** to other approaches.
- **Robotic vesicoscopy** is described and may grow, but the **8-mm ports of the Si/Xi** limit it.

Concept — The approaches **trade working space against the peritoneum**: transperitoneal buys space and familiarity by entering it; retroperitoneal and vesicoscopic preserve it at the cost of space — a trade that matters most where the peritoneum is needed, as in peritoneal dialysis.

BENEFITS OF MINIMALLY INVASIVE SURGERY

- Paediatric MIS benefits remain **debated** despite widespread robotic use.

Pain

- Smaller ports are **believed** to reduce pain. Robotic versus open pyeloplasty: one study found **less narcotic use**, another **no difference**.
- **No randomised trials**; most supportive data come from **adult robotic versus open prostatectomy**. Opioid-free analgesia is feasible after robotic prostatectomy, but **no paediatric MIS data** exist.

Admission / Length of Stay

- **Length of stay (LOS)** is a surrogate for recovery and pain; **enhanced recovery after surgery (ERAS)** is a separate field, but approach also shapes convalescence.
- **Pyeloplasty** meta-analyses: **MIS shorter than open**, and **robotic shorter than both laparoscopic and open**. **Ureteric reimplantation** meta-analysis: **shorter LOS** with MIS.
- Gains are in **hours** for pyeloplasty and reimplantation, but **days** for robotic **Mitrofanoff and augmentation cystoplasty**, despite longer operative times. MIS is a **recommended consideration in paediatric urology ERAS**.

Cosmesis

- Secondary, but scars last **for life**. **Incisions under 14.8 mm contract**; larger ones **enlarge** — favouring ports over open incisions, which usually exceed **20 mm**.
- Pursuit of fewer scars: **smallest robotic ports**; "scarless" laparoscopy through stab incisions without ports.
- Parents are **more satisfied after robotic than open pyeloplasty**.

THREE- VERSUS FOUR-PORT COSMESIS

Parents	Urologists
Prefer three ports	No difference in scarring

- **Single-site laparoscopy** improves cosmesis but is adopted slowly for technical difficulty; relocating ports, as in **HiDES**, improves satisfaction.

Pelvic Access

- As robotic prostatectomy showed, the robot has **revolutionised surgery in the confined pelvis**; it is feasible and potentially advantageous for **prostatic utricle, urogenital sinus, bladder neck reconstruction and distal ureteric pathology**.

Concept — The benefits measured objectively — **length of stay and scar** — are consistent; the benefit most often assumed, **pain**, is the least proven and largely borrowed from adults.

DRAWBACKS OF MINIMALLY INVASIVE SURGERY

- **Cost**: a da Vinci Xi with dual consoles exceeds **US\$2 million**, plus annual service — potentially prohibitive for a paediatric hospital; competition may lower prices.

COST STUDIES: PHIS VERSUS KID

Pediatric Health Information System (PHIS)	Kids' Inpatient Database (KID)
All robotic versus open cases: robotic significantly dearer	Pyeloplasty: robotic dearer, no significant overall difference

- Cost falls with **shorter operative time, higher hospital volume and shorter LOS**.
- **Teaching**: usually **one surgeon operates**; haptic feedback and real-time teaching are harder to reproduce.
- **Learning curve**: fellows needed about **37 robotic pyeloplasties** to match the attending's operative time; for experienced attendings the curve was less marked, the robot outperforming complex laparoscopic reconstruction. In a porcine model with novices, the robot beat laparoscopy on **operative time and anastomotic leak pressure**.
- The advantage is thus shown **against laparoscopy, not open surgery**; residents now train more robotically, so older curves may not apply.

KEY POINTS: ADVANTAGES AND DRAWBACKS OF MINIMALLY INVASIVE SURGERY

- Smaller ports may help **perioperative and postoperative pain control**.
- MIS is associated with **decreased LOS**.
- **Cosmesis is improved** compared with open incisions.
- MIS may help in **hard-to-access areas such as the deep pelvis**.
- **Robotic surgery costs more** than laparoscopic and open techniques.

Concept — The robot's drawbacks are **institutional** — **capital cost and training** — rather than clinical, and both diminish with volume.

COMPLICATIONS

Access Complications

- Entry risks **vascular injury, bowel injury, port-site hernia and subcutaneous emphysema**; Hasson and Veress complication rates do not differ.
- **Preperitoneal placement**: tip preperitoneal → **high opening pressure** → CO₂ forced preperitoneally → **subcutaneous emphysema**. Both are uncommon; **surgeon experience is the greatest predictor** of access complications.

Vascular and Visceral Complications

- **Major complications are rare**: vascular injury, small bowel injury, bleeding needing conversion, bladder perforation and vas deferens injury are all uncommon. Surgeons doing **more than 12 cases a year** have **fewer complications**; rates mirror adult urology.

MINOR VERSUS MAJOR VASCULAR INJURY

Commonest minor	Commonest major
Inferior epigastric , at port placement; avoid by direct visualisation in thin children	Right common iliac , lying directly behind the umbilicus

- Early recognition, treatment and reporting are paramount.
- **Pyeloplasty**, the commonest paediatric MIS procedure: **no difference in complications** between open, laparoscopic and robotic, even by severity; in **infants, no difference in complications or success**, MIS versus open.

Port-site hernia

- Rate **0.5–3%**, comparable to adults, who close ports by size and trocar type.

PORT-SITE HERNIA: FIRST PAEDIATRIC SERIES VERSUS ROBOTIC-PORT SERIES

First paediatric series	Robotic-port series
Upper end; risk higher around 1 year ; no difference with or without fascial closure	Lower end; 12- and 8-mm trocars, fascia closed at every site

- Hence **close port sites, especially in younger children**.

KEY POINTS: COMPLICATIONS

- Access complications: **vascular injury, bowel injury, port-site hernia, subcutaneous emphysema**.
- **Inferior epigastric** injury is the commonest minor vascular injury; the **right common iliac** is the commonest major vessel injured, being directly posterior to the umbilicus.
- Attend to **opening pressures** and **aspirate the Veress** on entry.

- Vascular and bowel injuries are **rare but potentially morbid**.
- Port-site hernia risk is **higher in smaller children**; **routine fascial closure is recommended**.

Concept — Complications cluster at **entry and exit** — the first needle or trocar and the fascial defect — so prevention rests on confirming position at entry, closing fascia at exit, and surgeon experience and volume.

TRAINING AND EDUCATION

- Just over half of practising paediatric urologists operate robotically; the main barrier is **lack of formal training**. Use is associated with **years in practice, formal robotic training and robot access**.
- **Residency training is variable**: most recent graduates felt **unable to operate robotically unsupervised** and lacked confidence in robotic pyeloplasty. Resident robotic cases rose, **mostly oncological**; **paediatric cases stayed overwhelmingly open**. Fellows' MIS volumes vary widely, and despite minimum case numbers there are **no formal MIS training guidelines**.
- **2015 international consensus curriculum**:

ROBOTIC CURRICULUM: TECHNICAL VERSUS NON-TECHNICAL SKILLS

Technical	Non-technical
Knots, suturing, transfer and dissection, camera use, arm manoeuvring, wrist articulation	Troubleshooting, patient selection, preparation, positioning, procedural steps, complication management

- **Sequence**: online modules → simulation (dry lab, cadaver or animal lab, virtual reality) → **supervised operating** → **full-length video** submitted for review. It has **not been formally adopted**.

Concept — Robotic adoption **follows training rather than evidence**; the deficit is not the absence of a curriculum, which has existed since 2015, but its adoption.

ERGONOMICS

Laparoscopic Ergonomics

- Laparoscopy causes **work-related musculoskeletal disorders**: the **fulcrum effect** (instruments pivot at the abdominal wall) and hands in **supination and ulnar deviation**. Most urologists report musculoskeletal pain, mostly ergonomic; **thumb pain tracks laparoscopic time**.
- Poor screen position, bed height and arm position cause fatigue and injury; teach ergonomics and a **neutral stance**. Port placement must balance triangulation with arm position; height, arm span and hand size matter.
- **Women are disproportionately affected**; **smaller hands** increase hand injury and instrument difficulty. Instruments for varied hand sizes and **core and back strengthening** help.

Robotic Ergonomics

- Seated at an **adjustable console** (office-chair standards apply), with **adjustable foot pedals** and a **saved ergonomic profile**.
- Surgeons comfortable with the **clutch** report **less discomfort**; poor clutch use leaves shoulders abducted and **elbows lifted off the armrest**.
- Motion tracking and electromyography: the robot causes **less upper-limb strain than laparoscopy**, but **not for the neck**, which is held more statically.

LAPAROSCOPIC VERSUS ROBOTIC ERGONOMIC TARGETS

	Laparoscopy (standing)	Robotic console (seated)
Work height	Table at 70–80% of elbow height	Locking chair, adjustable height
Neck	Flexion no more than 25°	Head flexion in viewer no more than 30°
Shoulders / arms	Retroversion 15–30° , blades back; arms about 90° at the side	Elbows 90–120° on the armrest
Vision	Monitor directly ahead at eye height , in the line of vision	In the viewer
Trunk / legs	Neutral stance	Lumbar support ; knees about a right angle
Main strain	Hands, wrists, thumb, shoulders	Neck (static)

KEY POINTS: ERGONOMICS

- Surgeons should know how to **improve their ergonomics** during MIS.
- Adjusting **arm position, bed height and monitor** minimises laparoscopic strain.
- Individual **console adjustments** minimise discomfort.
- Robotic surgeons should master the **clutch** to keep arm position.

Concept — Laparoscopy loads the upper limb through the fulcrum and the wrist; the robot offloads the upper limb but fixes the neck — **the ergonomic burden moves rather than disappears.**

FUTURE DIRECTIONS

- Pyeloplasty and reimplantation are increasingly robotic — **robotic pyeloplasty use rose steeply each year** — while complex procedures lag.

Same-day discharge and drainage

- Most MIS reconstructions are **observed overnight**, but uncomplicated robotic renal and ureteric cases, with **minimal stents and no catheters or drains**, were discharged **the same day** with low emergency return and readmission — **routine overnight stay is not always necessary.**
- **Stentless robotic pyeloplasty** is feasible in a small literature, but **stenting and urethral catheter drainage remain the gold standard.**

Expanding indications

- **Augmentation cystoplasty** for neurogenic bladder raises **capacity and compliance**, protecting the upper tracts; it needs a bowel segment, often with a **catheterisable channel**. Usually open, sometimes a laparoscopic start with a small incision for extracorporeal suturing.
- **First paediatric robotic augmentation: 2008, da Vinci S, fully intracorporeal.** Barriers: **VP shunts with fear of bowel spillage**, intracorporeal bowel work, case volume, prior surgery, habitus, learning curve. Later reports: **feasible, equivalent, longer operative times**; still performed by few.
- **Deep pelvis:** ureteric reimplantation is commonest; **robotic bladder neck reconstruction** for neurogenic bladder (first 2011), **utricle excision and urogenital sinus** repair also described.
- **Fetal surgery** is open (fetus exteriorised) or fetoscopic; fetoscopy shares laparoscopy's limits for fine repair such as **myelomeningocele closure**. Robotic fetal surgery is **animal-only**; **port size** is a main barrier.

Single-port surgery

- **da Vinci SP approved for urology in 2018**, the first formal robotic single-site platform: **2.5-cm cannula**, up to **three multijointed wristed instruments plus a wristed 3D high-definition camera**, manoeuvring to **restore triangulation**.
- **First paediatric SP series (2023)**: pyeloplasties for **pelviureteric junction obstruction** and one **Mitrofanoff**.
- **Limitation: at least 10 cm from port to target** is needed to deploy the instruments — prohibitive in small patients. Workarounds: **HiDES Pfannenstiel port** to maximise distance; a **gel-port to "float" the cannula** outside the body with sterile continuity. SP versus multiport pyeloplasty: **equivalent outcomes, longer operative time**.

Competing platforms

- da Vinci predominates; competitors could accelerate innovation. Smaller lenses and instruments would help **infants**, and the **2.5-cm SP cannula is too large** for many; paediatrics' **small market share** makes economically driven change less likely.

ROBOTIC PLATFORMS DESCRIBED IN THE TEXT

	da Vinci Xi	da Vinci SP	Senhance (Asensus)	Hugo RAS (Medtronic)
Access	Multiport, up to 4 arms	One 2.5-cm cannula	Multiport	Separate cart per arm
Smallest size	8 mm (no 5 mm)	25-mm port	3-mm instruments	—
Wrist / DOF	Wristed, 7 DOF	Multijointed, wristed	Laparoscopic-type; lacks rotational 7 DOF	Independent arms, 6 joints each , better range
Console	Closed	Closed	—	Open , 3D glasses, pistol-grip controllers
Status / use	Predominant	Urology approval 2018	—	Europe, Canada, Japan; U.S. Food and Drug Administration (FDA) investigational device exemption ; radical prostatectomy and partial nephrectomy feasible; paediatric data awaited

TABLE 39.1 FUTURE ROBOTIC TECHNOLOGY

Status	System (developer)	Approach	Camera	Arms	DOF	Haptics	Feature
Available (U.S.)	da Vinci (Intuitive)	Lap, LESS	HD-3D	4	7	None	Tremor filtration
Available (U.S.)	Flex (Medrobotics)	Transoral	HD-2D, semirigid or flexible	2	180° articulation	None	Steerable scope rigid once positioned
Available (not U.S.)	Mirosurge (German Aerospace Center)	Lap	HD-3D	3–5	7	Present	Adaptable MIRO arms
Available (not U.S.)	ViaCath (BIOTRONIK)	NOTES	—	1	9	None	Ureteroscopy and endovascular use
FDA anticipated	Senhance (TransEnterix)	Lap	HD-3D, eye-tracking	3	7	Present	Navigation; individual carts

Status	System (developer)	Approach	Camera	Arms	DOF	Haptics	Feature
FDA validation	Versius (Cambridge Medical Robotics)	Lap	HD-3D	Up to 5, modular	7	Present	Force and position sensed >1000 times per second; lightweight
FDA pending	SPORT (Titan Medical)	LESS	HD-3D	1	Multiple	None	Single incision; single-arm cart
FDA denied	SurgiBot (TransEnterix)	LESS	HD-3D	2	6	None	Internal triangulation; marketed in China
Clinical trial	MASTER (Nanyang Technological University)	NOTES	2D endoscope	2	9	Present	Smaller instruments, larger forces; navigation
Development	Hugo (Medtronic)	Lap	—	—	—	—	Mass utilisation to cut cost
Development	Hominis (Memic)	Lap, LESS, NOTES	—	Humanoid	360°	—	Humanoid-shaped arms
Development	MIVR miniature in vivo robot (Virtual Incision)	Advanced	HD, flexible tip	2	6	None	Artificial intelligence and machine learning
Development	Verb Surgical (J&J / Alphabet)	Advanced	—	—	—	—	"Surgery 4.0": robotics with machine learning

Lap, laparoscopic; LESS, laparo-endoscopic single-site surgery; NOTES, natural orifice transluminal endoscopic surgery; HD, high-definition; 2D/3D, two-/three-dimensional. Rebuilt and grouped by regulatory status.

- **Purpose:** maps the competitive field that should drive miniaturisation and lower cost.
- **Pattern:** rivals differ mainly by **haptic feedback** (absent in da Vinci; present in Senhance, Versius, Mirosurge, MASTER), **modular or separate-cart arms**, and **9 DOF** in NOTES platforms.
- **Table versus text:** the table lists **Hugo** as "**development**"; the text, being more recent, describes it as available outside the U.S. under FDA investigational exemption. Senhance's developer **TransEnterix** in the table is **Asensus** in the text — the same company, renamed.
- **Trap:** market-leading da Vinci has **7 DOF but no haptic feedback**; **SurgiBot** is the one denied by the FDA.

KEY POINTS: FUTURE CONSIDERATIONS

- Comfort with robotic platforms will **expand indications** and complex robotic procedures.
- Smaller robotic ports and instruments may aid **fetoscopic fetal surgery**.
- **New single-port platforms** have been used in paediatric patients.
- New platforms will increase **innovation and market competition**.

Concept — The limit on paediatric robotics is **the size of the hardware, not the capability of the operation**; progress depends on miniaturisation, which a small paediatric market cannot drive alone — hence the hope placed in competition.